Benchmarking

MaratonIC

Aqui estão as aplicações estudadas no processo de benchmarking:

1 UsacoGuide

1.1 Features

- Direcionamento para exercícios fora do site.
- Design da UI para escolher os exercícios.

1.2 Pontos negativos

- Não é possível testar soluções dentro do site.

2 TryHackMe

2.1 Features

- Vizualização intuitiva de aulas e exercícios competitivos.
- Design do site como todo. Como TryHackMe também é um site de competição (nesse caso de cybersecurity) e educativo, sua UI é uma boa inspiração para o site do MaratonIC. Além disso, o TryHackMe usa BootStrap Icons em seu site, que são ícones sem direitos autorais e podem ser facilmente copiados.

3 CodeForces

3.1 Features

- Vizualização de editoriais de soluções de problemas.
- Criação de editoriais de soluções de problemas escritos em Markdown.
- Participar de chats de discussão.
- Vizualização de notícias na página inicial.

4 League Of Legends

4.1 Features

- Ranking pessoal para estimular a competição entre os participantes. Também pode ser usado para ranquear problemas.
- Design de cores para diferenciar ranking, o que motiva as pessoas em competições.

5 Imagens

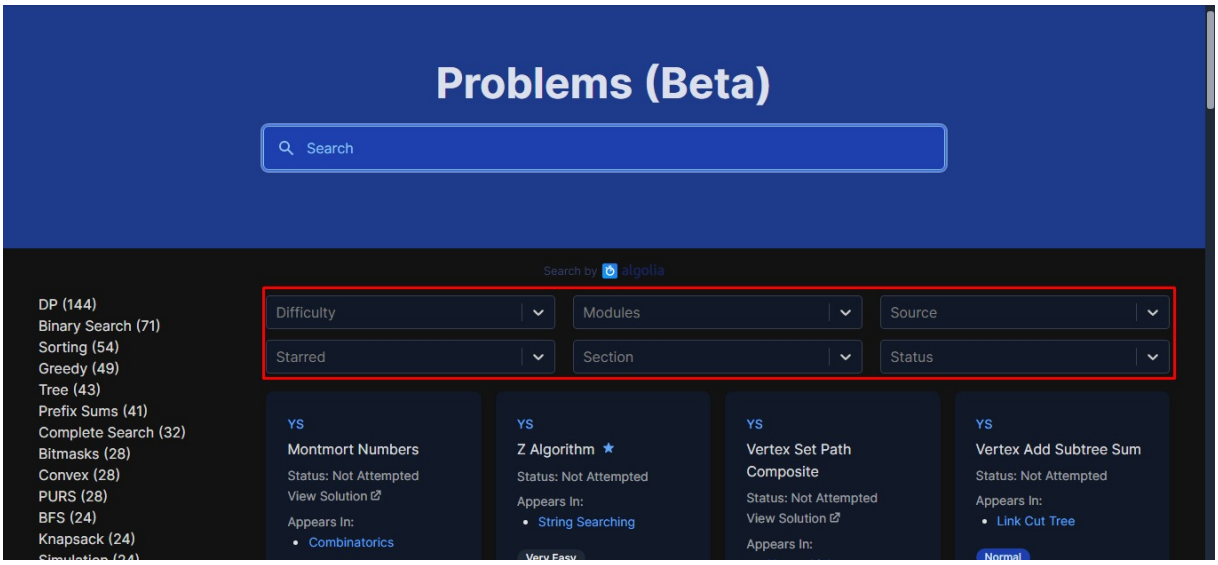


Imagem 1: UI do seletor de problemas da Usaco Guide

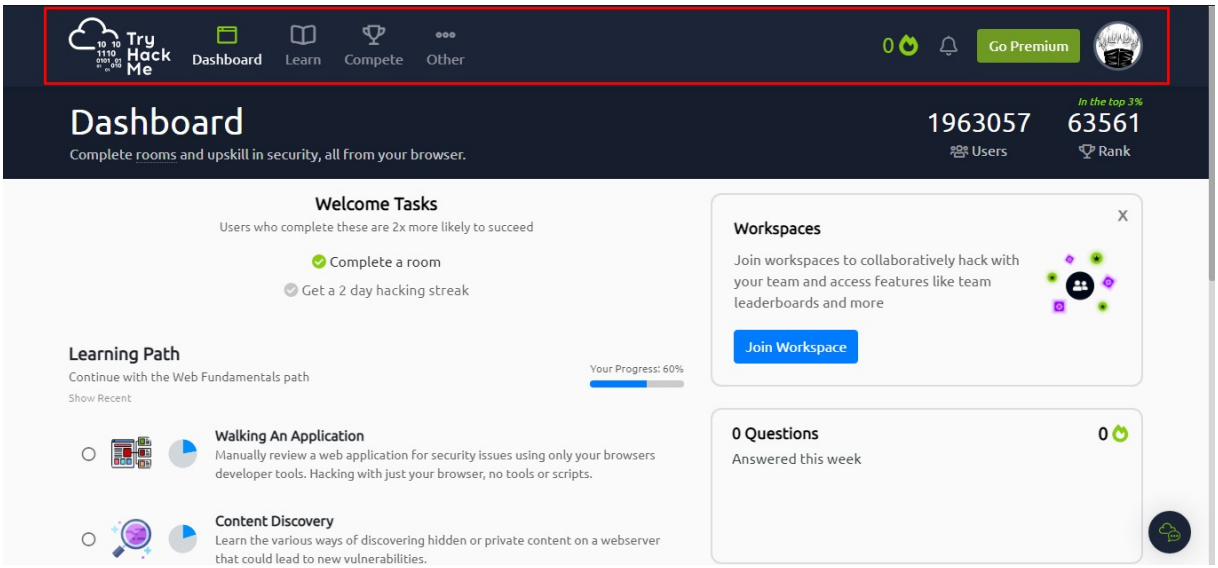


Imagem 2: Barra de navegação do TryHackMe

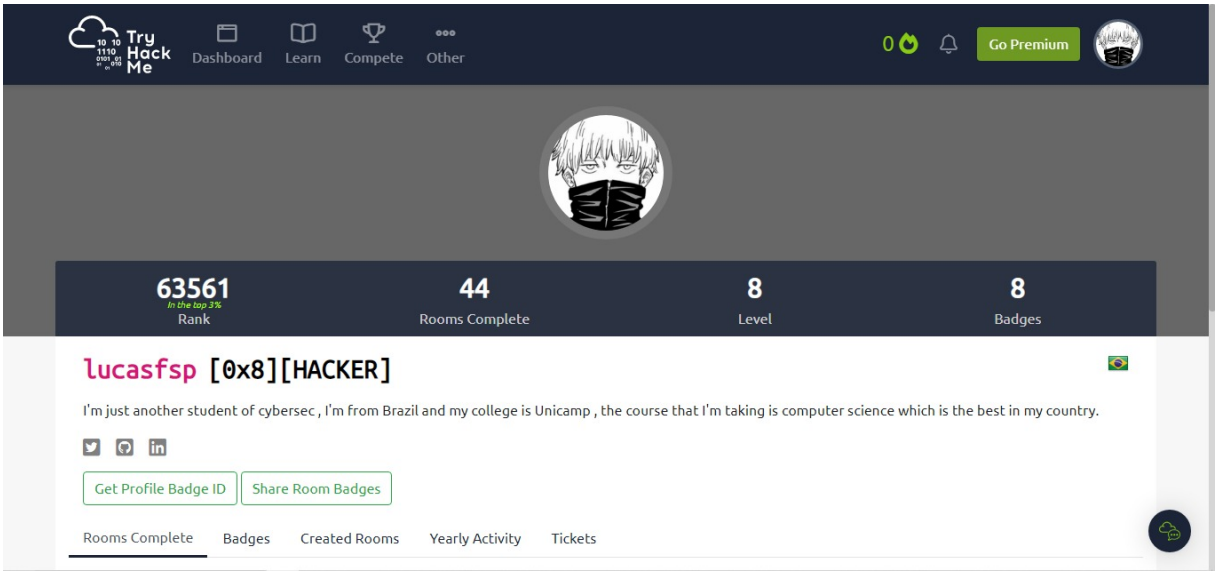
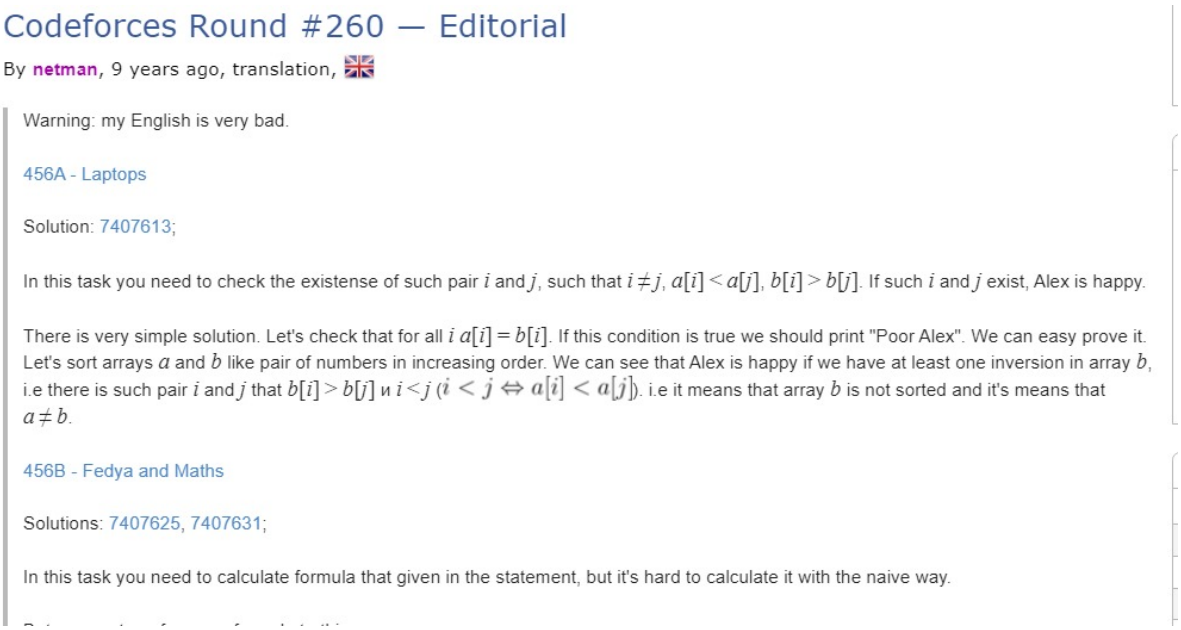


Imagem 3: Perfil de usuário do TryHackMe



Codeforces Round #260 — Editorial

By **netman**, 9 years ago, translation,

Warning: my English is very bad.

456A - Laptops

Solution: 7407613;

In this task you need to check the existense of such pair i and j, such that $i \neq j$, $a[i] < a[j]$, $b[i] > b[j]$. If such i and j exist, Alex is happy.

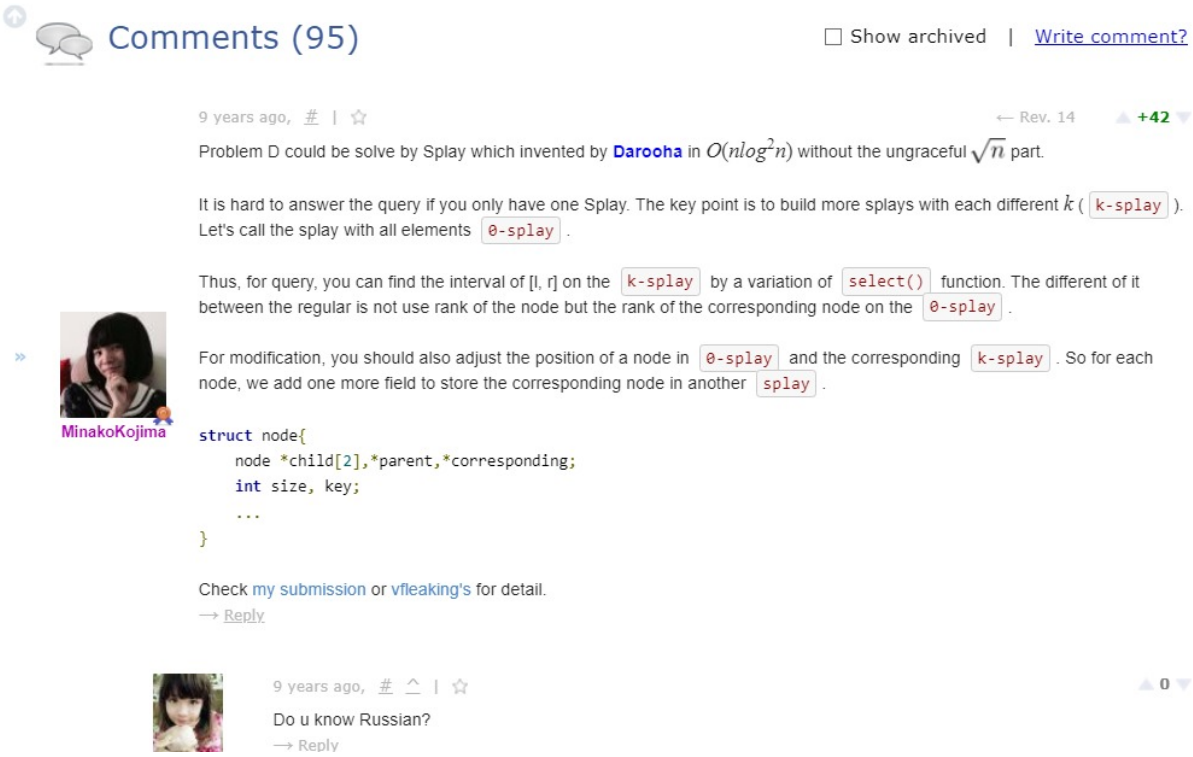
There is very simple solution. Let's check that for all i $a[i] = b[i]$. If this condition is true we should print "Poor Alex". We can easy prove it. Let's sort arrays a and b like pair of numbers in increasing order. We can see that Alex is happy if we have at least one inversion in array b, i.e there is such pair i and j that $b[i] > b[j]$ и $i < j$ ($i < j \Leftrightarrow a[i] < a[j]$). i.e it means that array b is not sorted and it's means that $a \neq b$.

456B - Fedya and Maths

Solutions: 7407625, 7407631;

In this task you need to calculate formula that given in the statement, but it's hard to calculate it with the naive way.

Imagem 4: Editorial de um problema no CodeForces



Comments (95)

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9 years ago, # | ☆ ← Rev. 14 +42

Problem D could be solve by Splay which invented by **Darooha** in $O(nlog^2n)$ without the ungraceful $\sqrt{n}$ part.

It is hard to answer the query if you only have one Splay. The key point is to build more splays with each different k (`k-splay`). Let's call the splay with all elements `0-splay` .

Thus, for query, you can find the interval of [l, r] on the `k-splay` by a variation of `select()` function. The different of it between the regular is not use rank of the node but the rank of the corresponding node on the `0-splay` .

MinakoKojima

For modification, you should also adjust the position of a node in `0-splay` and the corresponding `k-splay` . So for each node, we add one more field to store the corresponding node in another `splay` .

```
struct node{
    node *child[2],*parent,*corresponding;
    int size, key;
    ...
}
```

Check my submission or vfleaking's for detail.

→ Reply

9 years ago, # ^ | ☆ 0

Do u know Russian?

→ Reply

Imagem 5: Chat de discussão no CodeForces

Imagem 6: Sistema de ranking no League of Legends